

SOL SUSANA SANTOYO OLIVERA

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EDUCATION

Master's in mathematics

Ongoing: January 2025 - December 2026

Universidad de los Andes, Bogotá.

Master's thesis (in progress):

- Axiomatizing t -henselianity and augmenting fields.
What does the arithmetic of a field know about its possible henselian valuations?
- Monotilable amenable measure-preserving free actions on probability algebras
Model-theoretic properties of amenable ergodic theory

Undergraduate in mathematics, *cum laude*

January 2021 - December 2024

Universidad de los Andes, Bogotá.

Undergraduate thesis:

- Acciones de grupos amenables en álgebras de probabilidad
Una perspectiva de lógica continua en la teoría ergódica

RESEARCH INTEREST

I am interested in applications of model theory to other areas of mathematics, like algebra and analysis. For algebra, I am interested in valuation theory, ordered structures and algebraic objects with additional structure. In analysis, I study ergodic theory, functional analysis, and group actions on metric structures.

PROJECTS

These are the main projects I've been working on:

t -Henselian fields

research project

- We study fields admitting an elementary extension as a pure field that can be equipped with a henselian valuation. We are interested in how much of the valuation structure can be captured by the original pure field. From that framework, we study questions of classification (not independence property, strong dependence, dp-minimality) and definability (definable valuations, definable topologies).

This project is part of my master's thesis, supervised by professors Sylvie Anscombe (Université Paris Cité) and Pablo Cubides (Universidad de los Andes).

Measure-preserving actions

research project

- Given a measure-preserving action of a countable group on a standard Lebesgue space, we can see the induced action on the probability algebra as a metric structure. We study these theories from the perspective of continuous model theory. We study questions of classification (ω -categoricity, Schröder-Bernstein property, strong finiteness) and stability (non-forking, beautiful pairs, canonical bases) in these theories.

This project was also part of my master's thesis, supervised by professor Alexander Berenstein (Universidad de los Andes). Now it is work in progress for a paper, along with professor C. Ward Henson (University of Illinois Urbana-Champaign).

Multidimensional exact classes of vector spaces

research project

- Multidimensional exact classes are a common topic of study in finite and pseudofinite model-theory. They are classes of finite structures where there is a good control of size of definable sets. We study families of finite vector spaces, equipped with a linearly independent predicate, and other relational structures. We are working on connecting combinatorial notions (like measure) on the finite structures with geometric notions (Morley rank, SU rank) on their ultraproducts.

This project is ongoing work with professors Alexander Berenstein and Darío García at Universidad de los Andes.

TALKS AND PRESENTATIONS

Research Talks

- “Augmenting Fields by Ordered Abelian Groups,” Satellite event to the *Simposio Latinoamericano de Lógica Matemática*, Monguí 2026.
- “Groups of Automorphisms on Hilbert Spaces and Probability Algebras,” *Simposio Latinoamericano de Lógica Matemática*, Bogotá 2026.
- “Teoría de modelos de acciones que preservan la medida”
English translation: “Model Theory of Measure-Preserving Actions.” Congreso Colombiano de Matemáticas, Universidad Tecnológica de Pereira 2025.

Outreach Talks

- “El grafo aleatorio y juegos lógicos”
English translation: “The Random Graph and Logic Games.” Outreach Talk, Universidad de los Andes, Bogotá, 2023, 2024, 2026. Universidad Distrital, Bogotá, 2025
- “Teoría descriptiva de conjuntos en la teoría ergódica”
English translation: “Descriptive Set Theory in Ergodic Theory.” Legado Bourbaki, Interuniversity Undergraduate Seminar, Universidad Javeriana, Bogotá, 2023.
- “Cubos de Rubik y álgebra abstracta”
English translation: “Rubik’s Cubes and Abstract Algebra.” Outreach Talk for High School Students, Universidad de los Andes, Bogotá, 2022, 2023.

COURSES

These are some of the courses I’ve taken related to my research interests.

- Model theory
- Valued fields
- Hilbert’s Tenth Problem
- Commutative algebra
- Applications of stability to algebra (graduate tutorial course)
- Descriptive set theory
- Ergodic theory
- Functional analysis

Furthermore, I attend the model theory seminar at Universidad de los Andes regularly since fall 2024.

AWARDS AND ACADEMIC DISTINCTIONS

- **Préstamo Condonable para Carreras Especiales**, Universidad de los Andes, 2021–2025. Financial aid awarded on the basis of academic performance, available to students in selected degree programs.

- **Beca Yerly**, Department of Mathematics, Universidad de los Andes, 2022-02, 2023-2, 2024-2. Awarded for outstanding academic performance within the Department of Mathematics.
- **Beca Uniandinos**, Universidad de los Andes, 2023-01. University-wide academic merit scholarship.
- **Undergraduate in Mathematics, *cum laude***. Graduated with a GPA of 4.64/5.00, placing in the top 3% of the Faculty of Sciences over the preceding five years.
- **Master's in Mathematics**, current GPA: 4.89/5.00.

TEACHING

I was undergrad teaching assistant at the following courses, helping students and grading homework:

- Fall 2022: Honors differential calculus
- Spring 2023: Real analysis
- Fall 2023: Structural mathematics
- Spring 2024: Measure theory
Spring 2024: Abstract algebra 2
Spring 2024: Probability theory
- Fall 2024: Complex analysis

I have been teaching assistant at Universidad de los Andes for the following courses:

- Fall 2024: Linear algebra
- Spring 2025: Linear algebra
Spring 2025: Differential calculus
- Fall 2025: Linear algebra
Fall 2025: Integral calculus and differential equations
- Spring 2026: Linear algebra
Spring 2026: Vector calculus
- Fall 2026: Linear algebra
Fall 2026: Vector calculus

I have also worked at *Pentágono*, the center for math support for early STEM undergrad students since spring 2023.

LANGUAGES

Fluent in Spanish, English, and Italian. Basic German and ability to read in French.